# REDIAT TEREFE

Ph.D., Chemical Engineering | Heterogeneous Catalysis

rediat_ta@ch.iitr.ac.in | +91 745 209 9300 (India) | +251 910 604 091 (WhatsApp, Telegram) | rediatcheg.netlify.app (web)

## PROFILE

Ph.D. scholar in the Department of Chemical Engineering at IIT Roorkee, specializing in heterogeneous catalysis for converting biomass and C1 feedstocks into fuels and fuel intermediates. Experienced across the full catalyst workflow: design and synthesis of supported Ni and MOF-derived catalysts; characterization by XRD, BET, XPS, TPR/TPD/TPO, pulse chemisorption, FE-SEM and FTIR; catalytic testing; and product quantification by GC, GC-MS and HPLC to determine conversion, selectivity and yield. Seeking to apply this background to catalyst development for syngas-derived oxygenate fuels, such as oxymethylene ethers for sustainable aviation fuel.

## EDUCATION

**Ph.D., Chemical Engineering** *IIT Roorkee, India | Jul 2022 – Present | CGPA 8.3/10*

- Thesis title: "Production of furan intermediates from biomass waste and their upgrading to valuable chemicals using heterogeneous catalysts", expected to be completed in December 2026.

  Supervisor: Prof. Prakash Biswas, Department of Chemical Engineering, IIT Roorkee.

**M.Sc., Chemical Engineering** *Jimma University, Ethiopia | Graduated Feb 2021| CGPA 3.9/4*

- Thesis title: "Removal of Excess Free Fatty Acid from Edible Oil Using Betaine-Based Natural Deep Eutectic Solvents and Oil's Effect Analysis."

  Supervisor: Prof. Ramachandran K., Department of Petroleum Engineering, JCT Institutions.

**B.Sc., Chemical Engineering** *Jimma University, Ethiopia | Graduated June 2018 | CGPA 3.4/4*

- Final-year project title: "Bitumen Modification Using Waste Plastic (PET) for Road Bitumen Improvement and Minimization of Environmental Pollution."

  Supervisor: Prof. Muluken Eshetu, Department of Chemical Engineering, JIT, Jimma University.

## RESEARCH & TECHNICAL SKILLS

- **Catalyst synthesis & design:** Supported Ni catalysts, MOF-templated and N-doped Ni–alumina, magnetically separable catalysts; design of experiments (DOE) and reaction-parameter optimization by response surface methodology (RSM).
- **Catalytic testing:** Liquid-phase biomass conversion and transfer hydrogenation of oxygenates; collaborative gas-phase methane tri-reforming for syngas production and $CO_2$ conversion to formic acid; handling of $H_2$, $CH_4$ and $CO_2$ under laboratory safety procedures.
- **Characterization & analysis:** XRD, BET, TPR/TPD/TPO, pulse chemisorption, FE-SEM, FTIR, NMR, TGA, XPS; product quantification by GC, GC-MS and HPLC (conversion, selectivity, yield).
- **Computational & programming:** Python for data analysis and visualization; JavaScript/MERN-stack development; growing interest in AI/ML applications in catalysis and chemical engineering.
- **Software:** Origin, Design Expert, IBM SPSS, X'Pert HighScore, Edraw Max.
- **Languages:** English (professional working proficiency), Amharic (native), Gurage (native), Oromo (basic), Hindi (basic).

## PUBLICATIONS

- **Rediat T.**, Arisha A., Shubham S. D., Richa T., Sophie H., and Prakash B. (2026). "The conversion of furfural to furfuryl alcohol with high yield via catalytic hydrogenation transfer in the presence of nickel-embedded alumina catalyst derived from metal-organic framework template." *Catalysis Today*, https://doi.org/10.1016/j.cattod.2026.115883

- Arisha S., **Rediat T.**, Manas S., Richa T., Timothy S., Alessia T., Sophie H., and Prakash B. (2026). "*Amine-functionalized metal-organic framework-templated nitrogen-doped Ni-alumina catalyst for efficient methane tri-reforming.*" *Applied Materials Today*, https://doi.org/10.1016/j.apmt.2026.103292
- Arisha S., **Rediat T.,** and Prakash B. (2025). "*Optimization of reaction parameters by response surface methodology for the tri-reforming process over a Ni-silica catalyst to produce synthesis gas*." *International Journal of Hydrogen Energy*, https://doi.org/10.1016/j.ijhydene.2025.150378
- **Rediat T.** and Prakash B. (2025). "*The direct catalytic conversion of teff straw into fuel intermediate: Optimization of reaction parameters.*" *Biomass and Bioenergy*, https://doi.org/10.1016/j.biombioe.2025.108150
- **Rediat T**., Arisha S., and Prakash B. "*Transition-metal-free catalytic transfer hydrogenation of furfural to furfuryl alcohol over a pristine MIL-53(Al): reaction parameter optimization and kinetic insights.*" (Communicated).
- **Rediat T**., Arisha S., Manas S., and Prakash B. "*Catalyst-mediated valorization of biomass waste into furan intermediates using an Fe-catalyzed dual-phase system: An integrated experimental investigation and process optimization*" (Communicated).

## CONFERENCES & ACADEMIC TRAVEL

- Rediat Terefe and Prakash Biswas*, "*Catalytic hydrogenation of furfural to furfuryl alcohol using Al-supported Ni catalyst*". TICECEES-2024, International Chemical Engineering Conference on Energy, Environment and Sustainability, IIT Roorkee, Roorkee, India (poster presentation).
- Rediat Terefe and Prakash Biswas*, "*Enhancing the catalytic hydrogenation of furfural to furfuryl alcohol over Ni-loaded MIL-53: effect of calcination and reduction temperature on catalyst structure and activity*". EUROPACAT 2025, 16th European Congress on Catalysis, Trondheim, Norway (poster presentation).
- Rediat Terefe and Prakash Biswas*, "*Catalyst-mediated valorization of biomass waste into furan intermediates over Fe-catalyzed dual-phase system: An integrated experimental and optimization study*". CHEMCON 2026, 79th Annual session, Mumbai, India (stage presentation).

## HONORS & AWARDS

- Best Presentation Award, Research Scholars' Day, Department of Chemical Engineering, IIT Roorkee, Feb 2026.
- Certificate of Merit, 1st place, final year research competition, Department of Chemical Engineering, Jimma Institute of Technology, Jun 2018.
- Certificate of Recognition, Jimma University Community-Based Education Best Student Research Project, Annual Student Research Symposium, Jun 2018.

## INTEREST & ACTIVITIES

- Technology: designing systems and building them as working apps (MERN stack), including integrating AI models into practical tools.
- Sports: Basketball, Lawn tennis, and Football.
- Reading & learning: History and Theological books, Documentaries, New scientific publications, TED talks.

## REFERENCES

**Dr. Prakash Biswas**, Professor (Ph.D. supervisor), Department of Chemical Engineering, IIT Roorkee

prakash.biswas@ch.iitr.ac.in

**Dr. Basheshwer Prasad**, Professor, Department of Chemical Engineering, IIT Roorkee

basheshwer.prasad@ch.iitr.ac.in

**Dr. Muluken Eshetu**, Professor (B.Sc. project supervisor), School of Chemical Engineering, Jimma Institute of Technology, Ethiopia

mulushakached@gmail.com